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Problem 1. [5] Real numbers $x_1, x_2, \dots, x_{1434}$ satisfy $|x_n - x_{n+1}| = n^2$ for all integers $1 \leq n \leq 1433$. What is the sum of all possible values of $x_1 - x_{1434}$?

Problem 2. [5] A right triangle has hypotenuse of length 75 and a leg of length 45. A square has the same perimeter as the triangle. Find the area of the square.

Problem 3. [5] Let the 6-digit number $14A34B$ be divisible by 72. Find $10A + B$.

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Problem 4. [6] Pentagonal prisms M and N have the same height and have bases that are regular pentagons. The length of a side of the base of M is 6 times that of N . If the combined volumes is 999, find the volume of M .

Problem 5. [6] How many nonempty sets of the first 10 positive integers contain at most 2 prime numbers?

Problem 6. [6] Given that there is exactly one solution to the system of equations

$$\begin{aligned}y &= 3x^2 + 24x + c \\y &= 669,\end{aligned}$$

find c .

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Problem 7. [7] LeBron James enjoys playing basketball. For every 2-point shot LeBron takes, he has a $\frac{1}{2}$ chance of scoring and getting 2 points. For every 3-point shot LeBron takes, he has a $\frac{1}{3}$ chance of scoring and getting 3 points. LeBron has a score sequence that starts empty. For every 2-point shot he scores, he appends a 2 to the end of his score sequence, and for every 3-point shot he takes, he appends a 3 to the end of his score sequence. If LeBron ended up scoring 16 points, how many possible score sequences could LeBron have?

Problem 8. [7] Equilateral pentagon $ABCDE$ has side length 1, and $\angle E = \angle B = 90^\circ$. Compute the area of $ABCDE$.

Problem 9. [7] Let $f(n)$ represent the number of positive integer divisors of n . What is the minimum value of n that satisfies $f(f(f(n))) = 4$?

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Problem 10. [9] Let $x + \frac{2}{x} = 5$. Determine $x^2 + \frac{4}{x^2}$.

Problem 11. [9] Let $\triangle ABC$ have side lengths $AB = 5$, $AC = 7$, and $BC = 8$ with circumcenter O , and let D be the intersection of AO with its circumcircle. Find BD .

Problem 12. [9] Xook tells Xonk and Xoink to write a permutation of the word "STORY" uniformly at random. Xonk does this, but Xoink mishears the request and writes down a random permutation of the word "DIRTY". What is the probability that Xonk's and Xoink's permutations match in at least 2 spots? For example, if Xonk wrote "STORY" and Xoink wrote "DIRTY", then their permutations would match in 1 spot, namely the final Y.

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Problem 13. [10] How many positive integers n less than 10^4 have a digit a when written in base 10 such that a^2 and n have the same units digit?

Problem 14. [10] The real numbers x and y satisfy the system of equations

$$\begin{aligned}x^2 + 2xy + y^2 &= \frac{10}{x} + \frac{10}{y} \\ x^3 + y^3 &= 34.\end{aligned}$$

Compute $|x - y|$.

Problem 15. [10] Consider a 23×23 grid of unit squares. The single square located exactly at the center of the grid is removed.

Find the total number of squares (of all sizes, sides parallel to the grid) contained entirely within the grid that do not include the removed center square.

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Problem 16. [12] Let S be the set of positive integers such that c is in S if and only if there exist positive integers a, b such that $c = \gcd(a^b, b^a)$. Find the sum of the five smallest numbers in S .

Problem 17. [12] How many permutations $f(x)$ of the set $\{1, 2, 3, 4, 5, 6\}$ satisfy $f(f(f(f(x)))) = x$?

Problem 18. [12] Let $S = \sum_{k=1}^{\infty} \frac{\sin\left(\frac{4\pi k}{3}\right)}{2^k} = \frac{\sin\left(\frac{4\pi}{3}\right)}{2^1} + \frac{\sin\left(\frac{8\pi}{3}\right)}{2^2} + \dots$. Compute the value of S .

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Problem 19. [14] Call an integer n swell if $\frac{n^n + n!}{n + 1}$ is also an integer. Find the product of the first six positive swell integers.

Problem 20. [14] Thinking that AM-GM applies to negative real numbers, Jeff derives the (incorrect) inequality $27 - a - b \geq 9\sqrt[3]{ab}$ for all a, b . If we uniformly at random chose two not necessarily distinct integers a and b between 0 and 64, inclusive, what is the probability that they satisfy this inequality? You may find it helpful to reference the table below, which lists the number $\sqrt[3]{2}$ to $\sqrt[3]{7}$ to the nearest thousandth.

$\sqrt[3]{2}$	$\sqrt[3]{3}$	$\sqrt[3]{4}$	$\sqrt[3]{5}$	$\sqrt[3]{6}$	$\sqrt[3]{7}$
1.260	1.442	1.587	1.710	1.817	1.913

Problem 21. [14] 9 boys and 4 girls are seated randomly around a circular table. Compute the probability that no two girls sit next to each other.

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Problem 22. [17] In $\triangle ABC$, $\angle ABC = 82.5^\circ$, $AB = AC$, and $BC = \sqrt{6} - \sqrt{2}$. Point D lies inside ABC such that $\angle DBA = 7.5^\circ$ and $BD = CD$. If CD intersects AB at E , compute $\frac{1}{\sqrt{4 - AE \cdot EB}}$.

Problem 23. [17] Let $P(x)$ be the unique polynomial of minimal degree with coefficients in the set $\{0, 1, 2, 3, 4\}$ such that $(x^4 - x^3 + 2x^2 + 2x - 1)P(x)$ leaves the same remainder as x when divided by 5 for all positive integers x . Compute $P(10)$.

Problem 24. [17] A right circular cone has slant height 1. Compute the maximum possible volume of a rectangular prism inscribed within the cone.

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Problem 25. [20] Let p be the answer to problem 27. Sam starts with one stick of length 1. He begins to play a game with himself, where he randomly chooses a stick he currently has and breaks it perfectly in half, then chooses a new stick (possibly one of the halves) to break too, and so on. Given he breaks p sticks in total, find the expected value of the sum of the reciprocals of the lengths of the sticks. In other words, if the sticks after p breaks are of lengths (not necessarily distinct) $s_1, s_2, \dots, s_k$ then find the expected value of $\frac{1}{s_1} + \frac{1}{s_2} + \dots + \frac{1}{s_k}$.

Problem 26. [20] Let q be the answer to problem 25. Let r be a real number, and let S be the set of ordered pairs of positive real numbers (x, y) such that $\sqrt{x} + \sqrt{y} = r$. The minimum possible value of $\sqrt{x+49} + \sqrt{y+64}$ across all ordered pairs in S is $\sqrt{q}$. Compute r .

Problem 27. [20] Let r be the answer to problem 26. In triangle ABC with $AC > AB$, let ω be the circumcircle. Point D lies on ω such that $AD \perp BC$. Let ℓ_1 be the line AD , ℓ_2 be the line BD , and let ℓ_3 be the reflection of ℓ_2 across ℓ_1 . If the radius of ω is r , and ℓ_3 intersects ω at a point $E \neq D$ such that $CE = 12$, what is the length of AD ?